Interpreting Powers of i and the Complex Plane
Answer Key

Algebra 2 • Guided Concept Practice

Quick Answers

1. i
2. $-i$
3. $2+6i$
4. $-4+3i$
5. 5
6. $3i$, $-3i$
7. $2-3i$
8. $1+2i$
9. 2, -2, $2i$, $-2i$
10. $2+3i$, $2-3i$

Curriculum

Level: Algebra 2 • Guided Concept Practice
HSA-APR.B.3 – *problem 9*
HSA-REI.B.4 – *problems 6, 10*
HSN-CN – *problems 1, 2, 3, 4, 5, 7, 8*
Difficulty 1–5 of 5 across 10 tagged checks.

1. The three empty cells are $i^3 = i^2 \cdot i = -i$, $i^4 = (i^2)^2 = 1$, and $i^5 = i^4 \cdot i = i$. Since $i^4 = 1$, multiplying by i^4 changes nothing, so the list $i,\ -1,\ -i,\ 1$ repeats forever in blocks of four. That is why i^5 is back at the start: $\boxed{i^5 = i}$
.

2. Divide the exponent by the length of the cycle: $27 \div 4 = 6$ remainder 3. Six complete cycles land back at 1, and the remainder says take three more steps: $i^{27} = (i^4)^6 \cdot i^3 = 1 \cdot i^3$, so $\boxed{i^{27} = -i}$
.

The remainder is the whole method: remainder $0 \to 1$, $1 \to i$, $2 \to -1$, $3 \to -i$.

3. The arrows are $(3,2)$ and $(-1,4)$. Sliding the second arrow to the tip of the first moves it right -1 and up 4 from the point $(3,2)$, landing at $(2,6)$. In notation that is exactly adding the parts separately: $(3+2i)+(-1+4i) = (3-1)+(2+4)i$, so $\boxed{2+6i}$
.

The picture and the algebra are the same statement: complex addition is vector addition, componentwise.

4. $i(3+4i) = 3i + 4i^2 = 3i + 4(-1) = -4 + 3i$, so $\boxed{-4+3i}$
.

On the grid the arrow to $(3,4)$ becomes the arrow to $(-4,3)$: same length, turned a quarter turn ($90°$) counterclockwise about the origin. Accept any sentence saying the arrow rotated $90°$ counterclockwise without changing length. That is what multiplying by i *means* geometrically, and it explains the cycle in problem 1: four quarter turns return you to where you started, which is $i^4 = 1$.

5. The point $(3,4)$ sits 3 across and 4 up, so the right triangle dropped to the real axis has legs 3 and 4. By the Pythagorean theorem the distance from the origin is $\sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25}$, giving $\boxed{5}$
.

This distance is called the modulus, written $|3+4i|$.

6. $x^2 + 9 = 0$ gives $x^2 = -9$, so $x = \pm\sqrt{-9} = \pm 3i$.

$$\boxed{x = 3i \quad \text{or} \quad x = -3i}$$

Both solutions sit on the imaginary axis, at $(0,3)$ and $(0,-3)$: same distance from the origin, opposite directions. The parabola $y = x^2+9$ never crosses the x-axis precisely because neither solution has a real part to land on — the solutions exist, but not on the real number line the graph is drawn over.

7. Reduce each power with the remainder rule. $15 \div 4$ leaves remainder 3, so $i^{15} = -i$. $22 \div 4$ leaves remainder 2, so $i^{22} = -1$. Substituting: $3i^{15} - 2i^{22} = 3(-i) - 2(-1) = -3i + 2$, so $\boxed{2-3i}$
.

Watch the second sign: subtracting -1 adds, which is where this problem is usually lost.

8. $i^4 = 1$, so $i^4(1+2i) = 1(1+2i)$, giving $\boxed{1+2i}$
— you land exactly where you started.

The picture says it first: each multiplication by i turns the arrow a quarter turn, so four of them is a full $360°$ turn and every point returns to itself. The arithmetic only confirms what the rotation already guaranteed.

9. Factor as a difference of squares twice: $x^4 - 16 = (x^2-4)(x^2+4) = (x-2)(x+2)(x^2+4)$. The first two factors give real zeros; $x^2 + 4 = 0$ gives $x^2 = -4$, so $x = \pm 2i$.

$$\boxed{x = 2,\ -2,\ 2i,\ -2i}$$

The real zeros are 2 and -2; the non-real ones are $2i$ and $-2i$. Plotted, the four points sit at the four compass points of a circle of radius 2 about the origin — one every quarter turn, which is the same fourfold symmetry the powers of i have.

10. Completing the square: $x^2 - 4x + 13 = (x-2)^2 + 9 = 0$, so $(x-2)^2 = -9$ and $x - 2 = \pm 3i$.

$$\boxed{x = 2 + 3i \quad \text{or} \quad x = 2 - 3i}$$

(The quadratic formula gives the same thing: the discriminant is $16 - 52 = -36$, and $\sqrt{-36} = 6i$.)

The two points are $(2, 3)$ and $(2, -3)$ — reflections of each other across the real axis. This must happen whenever the coefficients are real: the quadratic formula adds and subtracts the same $\sqrt{\text{negative}}$ term, so the two answers share a real part and carry opposite imaginary parts. Non-real solutions of a real polynomial always come in such conjugate pairs, which is also why problem 9 produced $2i$ and $-2i$ together.